Math 225A
Topics in Number Theory

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Introduction

This note is based on the lecture of Professor Krishna, and written by Nayeong Kim. In this course, we will explore the following topics.

- Galois Cohomology
- Central Simple Algebras
- Brauer Groups and their Representation with Galois Cohomology
- Brauer Groups of Local and Global Fields

AI Disclaimer

I am using Claude for maintaining the notes in the git repository and for generating formatting and some Latex code.

1 Group Cohomology

Let G be a discrete group, and let $p\colon EG \to BG$ be the universal principal G-bundle over the classifying space BG. Its defining property is that, for every paracompact Hausdorff space X, pulling back along maps $X \to BG$ induces a bijection

$$[X, BG] \xrightarrow{\cong} \mathrm{Prin}_G(X)$$

between homotopy classes of maps $X \to BG$ and isomorphism classes of principal G-bundles over X.

In particular, let $\pi\colon P \to X$ be a principal G-bundle. Then there is a map $f\colon X \to BG$, unique up to homotopy, and a G-equivariant map $\tilde{f}\colon V \to EG$ such that the square

$$\begin{array}{ccc} V & \xrightarrow{\tilde{f}} & EG \\ {\scriptstyle\pi}\downarrow & & \downarrow{\scriptstyle p} \\ X & \xrightarrow[f]{} & BG \end{array}$$

is a pullback; that is, $V \cong f^*EG$ as principal G-bundles.

We note that since G is discrete, we have $\Pi_1(BG) \cong G$ and $\Pi_i(BG) = 0$ for $i > 1$. So BG is a $K(G, 1)$ space, and we have

$$H^i(G, M) \cong H^i(BG, M)$$

for any G-module M.

Now, when we take a look at $H^*_{sing}(BG, \mathbb{Z})$, it is much more complicated than BG but it gives us a lot of information about G. This gives a geometric motivation for group cohomology as we study representations of G to understand G better.

We can study it when we want to understand G, the Galois group of some field. For example, when $G = \mathbb{Z}$, we have $BG = S^1$ and $EG = \mathbb{R}$ and we can compute $H^*(S^1, \mathbb{Z})$. For another example, when $G = \mathbb{Z}/2\mathbb{Z}$, we have $BG = \mathbb{R}P^\infty$ and $EG = S^\infty$ and we can compute $H^*(\mathbb{R}P^\infty, \mathbb{Z})$. More generally, when $G = \mathbb{Z}/n\mathbb{Z}$, we have $BG = L^\infty_n := S^\infty/(\mathbb{Z}/n\mathbb{Z})$, the infinite lens space, and $EG = S^\infty$. Here we regard S^∞

as the unit sphere in $\mathbb{C}^\infty = \bigcup_k \mathbb{C}^k$, and a generator of $\mathbb{Z}/n\mathbb{Z}$ acts by multiplication by $\zeta = e^{2\pi i/n}$ on each coordinate. This action is free, and S^∞ is contractible, so $S^\infty \to L_n^\infty$ is a universal principal $\mathbb{Z}/n\mathbb{Z}$-bundle. Using the CW structure on L_n^∞ with one cell in each dimension, whose cellular chain complex is

$$\mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \mathbb{Z} \xleftarrow{n} \mathbb{Z} \xleftarrow{0} \cdots,$$

we obtain

$$H^k(L_n^\infty, \mathbb{Z}) \cong \begin{cases} \mathbb{Z} & k = 0, \\ \mathbb{Z}/n\mathbb{Z} & k > 0 \text{ even}, \\ 0 & k \text{ odd}. \end{cases}$$

In fact, as a ring, $H^*(L_n^\infty, \mathbb{Z}) \cong \mathbb{Z}[x]/(nx)$ with $\deg x = 2$. Setting $n = 2$ recovers $H^*(\mathbb{R}P^\infty, \mathbb{Z})$, since $L_2^\infty = \mathbb{R}P^\infty$.

To study such objects, we cannot find a good answer with topological methods, so we need to use algebraic methods. We can study cohomology groups and Galois cohomology groups with algebraic methods.

What is a group cohomology? For simplicity, we assume that G is a finite group (although the profiniteness is enough). Let A be an abelian group with a G-action, i.e., a G-module. Let G-Mod be the category of G-modules and G-equivariant homomorphisms. We note that G-Mod is an abelian category with enough injectives.

Let's consider an example. Let $\mathbb{Z}[G]$ be the group algebra of G over $\mathbb{Z}$. In fact, this is a canonical way of constructing a ring from a group. Here are some properties of $\mathbb{Z}[G]$.

Remark 1.

1. $\mathbb{Z}[G]$ is commutative if and only if G is abelian.
2. $\mathbb{Z}[G]$ is noetherian if and only if G is finitely generated.

Remark 2. The algebra $\mathbb{Z}[G]$ is an augmented $\mathbb{Z}$-algebra with an augmentation $\epsilon\colon \mathbb{Z}[G] \to \mathbb{Z}$ map given by

$$\sum_{g \in G} m_g g \longmapsto \sum_{g \in G} m_g.$$

The augmentation ideal I_G is defined as the kernel of ϵ so it is clear that it is a 2-sided ideal of $\mathbb{Z}[G]$.

We also note that $\mathbb{Z}[G]$ is a free $\mathbb{Z}$-module with a $\mathbb{Z}$-basis G. So is I_G with the following lemma.

Lemma 1. The ideal I_G has a $\mathbb{Z}$-basis $\{g-1 \mid g \in G, g \neq 1\}$.

Proof. Let $g \in G$ with $g \neq 1$. Then we have

$$\epsilon(g-1) = \epsilon(g) - \epsilon(1) = 1 - 1 = 0,$$

so $g - 1 \in I_G$. Now, let $\sum_{g\in G} m_g g \in I_G$. Then we have

$$0 = \epsilon\left(\sum_{g\in G} m_g g\right) = \sum_{g\in G} m_g.$$

So we can write

$$\sum_{g\in G} m_g g = m_1 + \sum_{g\neq 1} m_g g = -\sum_{g\neq 1} m_g + \sum_{g\neq 1} m_g g = \sum_{g\neq 1} m_g(g-1).$$

This shows that $\{g-1\}_{g\neq 1}$ generates I_G as a $\mathbb{Z}$-module.

To show that it is linearly independent, suppose that $\sum_{g\neq 1} m_g(g-1) = 0$. Then we have $\sum_{g\neq 1} m_g g + \sum_{g\neq 1}(-m_g)1 = 0$. Since the elements of G are linearly independent in $\mathbb{Z}[G]$, we must have $m_g = 0$ for all $g \neq 1$. □

The action of G is invariant on I_G since $h\cdot(g-1) = hg - h = (hg-1)-(h-1) \in I_G$ for all $g, h \in G$. So we have $\mathbb{Z}[G] = \mathbb{Z} \oplus I_G$ as G-modules.

As we mentioned, we can consider a ring $\mathbb{Z}[G]$ canonically from a group G. Not surprisingly, we can consider a G-module A as a $\mathbb{Z}[G]$-module via $(\sum_{g\in G} m_g g)\cdot a = \sum_{g\in G} m_g(g\cdot a)$ for $a \in A$. The converse is also true. There is a composition of homomorphism $G \to \mathbb{Z}[G]^\times \to \mathrm{Aut}_{ab}(A)$ which gives a canonical G-action on A.

Proposition 1. There is an equivalence of abelian categories between G-Mod and $\mathbb{Z}[G]$-Mod.

Now we consider a functor $F : G\text{-}\mathsf{Mod} \to \mathsf{Ab}$ which sends A to $A^G = \{a \in A \mid g \cdot a = a \text{ for all } g \in G\}$. We may consider an abelian group A as a G-module with a trivial action so we have a functor $\mathrm{tr} : \mathsf{Ab} \to G\text{-}\mathsf{Mod}$.

Claim 1. We claim that (tr, F) is an adjoint pair of functors.

Proof. For any $A \in \mathsf{Ab}$ and $B \in G\text{-}\mathsf{Mod}$, we have a natural isomorphism

$$\mathrm{Hom}_{G\text{-}\mathsf{Mod}}(\mathrm{tr}(A), B) \cong \mathrm{Hom}_{\mathsf{Ab}}(A, B^G).$$

For any G-equivariant homomorphism $\phi : \mathrm{tr}(A) \to B$, we have a homomorphism $\psi : A \to B^G$ given by

$$\psi(a) = \phi(a) \in B^G.$$

Conversely, for any homomorphism $\psi : A \to B^G$, we have a G-equivariant homomorphism $\phi : \mathrm{tr}(A) \to B$ given by

$$\phi(a) = \psi(a) \in B.$$

It is clear that these two maps are inverses of each other. □

With the claim, we have a corollary.

Corollary 1. The invariant functor $F : G\text{-}\mathsf{Mod} \to \mathsf{Ab}$ is left exact.

But note that F is not right exact in general. For example, let $G = \mathbb{Z}/2\mathbb{Z}$ and take the augmentation sequence $0 \to I_G \to \mathbb{Z}[G] \to \mathbb{Z} \to 0$ which is exact with the definition of I_G. But $0 \to I_G^G = 0 \to \mathbb{Z}[G]^G = \mathbb{Z} \to \mathbb{Z}^G = \mathbb{Z} \to 0$ is not exact.

With the left exactness of F, we can consider the right derived functor R^iF. But we need to be careful; we need to have enough injectives in $G\text{-}\mathsf{Mod}$ to define the derived functor.

Lemma 2. Let A be an abelian group. Then there exists an injective abelian group I and an injective homomorphism $A \to I$.

Proof. Let S be the set of all abelian group homomorphisms from A to $\mathbb{Q}/\mathbb{Z}$. Let I be S-copies of $\mathbb{Q}/\mathbb{Z}$. It is clear that I is injective since $\mathbb{Q}/\mathbb{Z}$ is injective.

We have a homomorphism $\Phi : A \to I$ given by $a \mapsto (\phi(a))_{\phi \in S}$. We want to show that Φ is injective, and it suffices to show that for any nonzero $a \in A$, there exists a homomorphism $\phi : A \to \mathbb{Q}/\mathbb{Z}$ such that $\phi(a) \neq 0$.

Let $a \in A$ be nonzero. Then we have a subgroup $\langle a \rangle$ of A generated by a. We have a homomorphism $\psi : \langle a \rangle \to \mathbb{Q}/\mathbb{Z}$ given by $ma \mapsto m/n + \mathbb{Z}$ where n is the order of a. By the injectivity of $\mathbb{Q}/\mathbb{Z}$, we can extend ψ to a homomorphism $\phi : A \to \mathbb{Q}/\mathbb{Z}$. Then we have a non-zero homomorphism ϕ with $\phi(a) = \psi(a) = 1/n + \mathbb{Z} \neq 0$. It follows that Φ is injective. □

Proposition 2. Let R be a ring and M be an R-module. Then there are enough injective R-modules.

Proof. Let M be a left R-module. Look at the composition of R-module homomorphisms $M \xrightarrow{\sim} \mathrm{Hom}_R(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, M)$ where $M \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Z}}(R, M)$ is given by $m \mapsto (r \mapsto rm)$. We take an injective module I as in the previous lemma. Then we have an injection $\mathrm{Hom}_{\mathbb{Z}}(R, M) \hookrightarrow \mathrm{Hom}_{\mathbb{Z}}(R, I)$ via $M \hookrightarrow I$.

It is left to show that $I' := \mathrm{Hom}_{\mathbb{Z}}(R, I)$ is injective as an R-module. Take any injection $A \hookrightarrow B$ of R-modules. We want to show that $\mathrm{Hom}_R(B, I') \twoheadrightarrow \mathrm{Hom}_R(A, I')$. We observe that $\mathrm{Hom}_R(B, I') = \mathrm{Hom}_R(B, \mathrm{Hom}_{\mathbb{Z}}(R, I)) \cong \mathrm{Hom}_{\mathbb{Z}}(B, I)$ via the adjointness of Hom functors. Similarly, we have $\mathrm{Hom}_R(A, I') \cong \mathrm{Hom}_{\mathbb{Z}}(A, I)$. Since I is injective as an abelian group (as a $\mathbb{Z}$-module), we have $\mathrm{Hom}_{\mathbb{Z}}(B, I) \twoheadrightarrow \mathrm{Hom}_{\mathbb{Z}}(A, I)$. It follows that $\mathrm{Hom}_R(B, I') \twoheadrightarrow \mathrm{Hom}_R(A, I')$ and I' is injective as an R-module. □

Now we work with a G-module A.

Lemma 3. $A^G = \mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, A)$.

Proof. By adjointness, we have $\mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, A) \cong \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}, A)^G = A^G$. □

2 Derived Categories

Since we have enough injective G-modules, we can consider the derived functor. We consider an injective resolution of A as a $\mathbb{Z}[G]$-module

$$0 \to A \to I^0 \to I^1 \to I^2 \to \cdots .$$

From the cochain complex, we have

$$\mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A) = H^i(\mathrm{Hom}_{\mathbb{Z}[G]}(\mathbb{Z}, I)).$$

We note that it doesn't depend on the choice of injective resolution of A; hence it is well-defined.

Definition 1. Define the i-th cohomology group of G with coefficients in A as $H^i(G, A) := \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A)$.

Here is a corollary of the previous lemma.

Corollary 2. We have $H^0(G, A) = A^G$.

How do we actually compute $H^i(G, A)$? The hardness of the computation depends on the choice of the injective resolution of A. For $\mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A)$, instead of taking an injective resolution of A, we can take a projective resolution of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module. One benefit of this approach is that projective resolutions are often easier to construct than injective resolutions since projective modules are direct summands of free modules.

Recall the following theorem.

Theorem 1. Let M be a projective R-module and N be an R-module. Then $\mathrm{Ext}^i_R(M, N) = 0$ for all $i > 0$.

Proof. We may assume that M is a free R-module since projective modules are direct summands of free modules. Let $M = \bigoplus_{j \in I} R$ be a free R-module. Then we have $\mathrm{Ext}^i_R(M, N) = \mathrm{Ext}^i_R\left(\bigoplus_{j \in I} R, N\right) = 0$ for all $i > 0$. □

Corollary 3. If $f : A \to B$ is a G-invariant map, then there is a canonical map $f^i : H^i(G, A) \to H^i(G, B)$ for all $i \geq 0$.

Corollary 4. $H^0(G, A) = A^G$, and $H^1(G, A)$ is the set of equivalence classes of extensions of $\mathbb{Z}$ by A as $\mathbb{Z}[G]$-modules.

Corollary 5. Let $0 \to A' \to A \to A'' \to 0$ be a short exact sequence of G-modules. Then we have a long exact sequence

$$\cdots \to H^i(G, A') \to H^i(G, A) \to H^i(G, A'') \to H^{i+1}(G, A') \to \cdots .$$

We are still working with a finite group G.

Definition 2. The *norm element* of $\mathbb{Z}[G]$ is defined as the sum of all elements in G:

$$N := \sum_{g \in G} g.$$

Remark 3.

1. The norm element N is a central element of $\mathbb{Z}[G]$ since we have

$$gN = g \sum_{h \in G} h = \sum_{h \in G} gh = N$$

for any $g \in G$. Likewise $Ng = N$. Hence $gN = N = Ng$.

2. N is a G-invariant element of $\mathbb{Z}[G]$ since we have

$$g \cdot N = g \cdot \sum_{h \in G} h = \sum_{h \in G} gh = N$$

for any $g \in G$.

Theorem 2. Let G be a finite group of order m and A be a G-module such that A is also an R-module where R is any commutative ring with $1/m \in R$. Then N/m is a central idempotent of $R[G]$. Moreover, $H^i(G, A) = 0$ for all $i > 0$ and $H^0(G, A) = eA$ where $e = N/m$.

Proof. We have

$$N^2 = \left(\sum_{g\in G} g\right) N = \sum_{g\in G} gN = \sum_{g\in G} N = mN.$$

Therefore,

$$(N/m)^2 = N^2/m^2 = mN/m^2 = N/m$$

i.e. N/m is an idempotent of $R[G]$. We also have N/m is central since N is central.

Then we have $R[G] = R[G]e \oplus R[G](1-e)$ where $e = N/m$ is an idempotent of $R[G]$. In particular, $R[G]e$ is a projective $R[G]$-module since it is a direct summand of a free module $R[G]$. It follows that $\mathrm{Ext}^i_{R[G]}(R[G]e, A) = 0$ for all $i > 0$.

Since $1/m \in R$, we have that A is naturally a $\mathbb{Z}[1/m][G]$-module. Let $P_\bullet \to \mathbb{Z}$ be a resolution of $\mathbb{Z}$ by free $\mathbb{Z}[G]$-modules. Localizing at m, we obtain $P_\bullet[1/m] \to \mathbb{Z}[1/m]$ which is also a free resolution of $\mathbb{Z}[1/m]$ by $\mathbb{Z}[1/m][G]$-modules.

Since m acts invertibly on A, the universal property of localization gives an isomorphism $\mathrm{Hom}_{\mathbb{Z}[1/m][G]}\big(P_i[1/m], A\big) \xrightarrow{\sim} \mathrm{Hom}_{\mathbb{Z}[G]}(P_i, A),\ f \mapsto f|_{P_i}$, which is natural in P_i: every $\mathbb{Z}[G]$-linear map $P_i \to A$ extends uniquely to $P_i[1/m]$ by $f(x/m^k) = m^{-k}f(x)$. These isomorphisms commute with the differentials, so

$$\begin{aligned} H^i(G, A) &= H^i\big(\mathrm{Hom}_{\mathbb{Z}[G]}(P_\bullet, A)\big) \\ &\cong H^i\big(\mathrm{Hom}_{\mathbb{Z}[1/m][G]}(P_\bullet[1/m], A)\big) \\ &= \mathrm{Ext}^i_{\mathbb{Z}[1/m][G]}(\mathbb{Z}[1/m], A). \end{aligned}$$

Applying the previous result with $R := \mathbb{Z}[1/m]$, we are done with $\mathbb{Z}[1/m] = R[G]e = \mathbb{Z}[1/m][G]e$ as $\mathbb{Z}[1/m][G]$-modules. Since $e = N/m$, we know $\mathbb{Z}[1/m][G]e = \mathbb{Z}[1/m]e$. By augmentation, we have $\mathbb{Z}[1/m]e \cong \mathbb{Z}[1/m]$ as $\mathbb{Z}[1/m][G]$-modules. Therefore, we have $H^i(G, A) = 0$ for all $i > 0$. For $i = 0$, we have

$$H^0(G, A) = \mathrm{Hom}_{\mathbb{Z}[1/m][G]}(\mathbb{Z}[1/m], A) = \mathrm{Hom}_{\mathbb{Z}[1/m][G]}(\mathbb{Z}[1/m][G]e, A) = eA.$$

□

Corollary 6. Let G be a finite group and A be a G-module which is also a $\mathbb{Z}[1/m]$-module where $m = |G|$. Then $H^i(G, A) = 0$ for all $i > 0$.

Corollary 7. Let G be a finite group. Then $H^0(G, \mathbb{Z}) = \mathbb{Z}, H^1(G, \mathbb{Z}) = 0$, and $H^i(G, \mathbb{Z}) = H^{i-1}(G, \mathbb{Q}/\mathbb{Z})$ for all $i > 1$.

Proof. We have an exact sequence of trivial G-modules $0 \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{Q}/\mathbb{Z} \to 0$. It gives a long exact sequence

$$0 \to H^0(G, \mathbb{Z}) \to H^0(G, \mathbb{Q}) \to H^0(G, \mathbb{Q}/\mathbb{Z}) \to H^1(G, \mathbb{Z}) \to H^1(G, \mathbb{Q}) \to \cdots$$

Since $H^i(G, \mathbb{Q}) = 0$ for all $i > 0$ by the previous corollary, we are done. □

3 Bar Resolution

Let G be any discrete group. We want to construct a free resolution of $\mathbb{Z}$ as $\mathbb{Z}[G]$-modules. We set $B_{-1} := \mathbb{Z}$ and the 0-th power of G is the trivial group. For $n \geq 0$, we define B_n to be the free $\mathbb{Z}[G]$-module on G^n. We define the following maps $d_n : B_n \to B_{n-1}$ for $n \geq 0$ (we want the maps to be boundary maps):

$$
\begin{aligned}
&d_0 = \epsilon \quad \text{(augmentation map)},\\
&d_n : B_n \to B_{n-1} \text{ which is defined as}\\
&\quad d_n = \sum_{i=0}^{n} (-1)^i d_n^i,\\
&\quad \text{where } d_n^0([g_1|\cdots|g_n]) = g_1 \cdot [g_2|\cdots|g_n],\\
&\qquad\qquad d_n^i([g_1|\cdots|g_n]) = [g_1|\cdots|g_ig_{i+1}|\cdots|g_n] \text{ for } 1 \leq i \leq n-1,\\
&\qquad\qquad d_n^n([g_1|\cdots|g_n]) = [g_1|\cdots|g_{n-1}].
\end{aligned}
$$

The maps d_n are boundary maps because of their simplicial structure (look at the Dold–Kan correspondence).

Theorem 3. The sequence $B_\bullet$ is exact.

Proof. We shall show that $\cdots \to B_n \xrightarrow{d_n} B_{n-1} \to \cdots \to B_0 \xrightarrow{d_0} B_{-1} \to 0$ is exact by constructing a null-homotopy. We want to construct maps $h_n : B_n \to B_{n+1}$ such that $d_{n+1} \circ h_n + h_{n-1} \circ d_n = \mathrm{id}_{B_n}$ for all $n \geq -1$. Define $h_n : B_n \to B_{n+1}$ as

$$h_n(g_0 \cdot [g_1|\cdots|g_n]) = [g_0|g_1|\cdots|g_n].$$

For $n = -1$, it is clear that $d_0 \circ h_{-1} = \mathrm{id}_{B_{-1}}$. Then when $n \geq 1$, we have

$$
\begin{aligned}
d_{n+1} \circ h_n([g_1|\cdots|g_n]) &= d_{n+1}([e|g_1|\cdots|g_n])\\
&= [g_1|\cdots|g_n] - [g_1|\cdots|g_n]\\
&\quad + \sum_{i=1}^{n-1} (-1)^{i+1}[e|g_1|\cdots|g_ig_{i+1}|\cdots|g_n] + (-1)^{n+1}[e|\cdots|g_{n-1}]
\end{aligned}
$$

and

$$\begin{aligned} h_{n-1} \circ d_n([g_1|\cdots|g_n]) &= h_{n-1}(g_1 \cdot [g_2|\cdots|g_n] \\ &\quad + \sum_{i=1}^{n-1}(-1)^i[g_1|\cdots|g_ig_{i+1}|\cdots|g_n] + (-1)^n[g_1|\cdots|g_{n-1}]) \\ &= [g_1|\cdots|g_n] + \sum_{i=1}^{n-1}(-1)^i[e|g_1|\cdots|g_ig_{i+1}|\cdots|g_n] \\ &\quad + (-1)^n[g_1|\cdots|g_{n-1}]. \end{aligned}$$

Therefore, we have

$$d_{n+1} \circ h_n([g_1|\cdots|g_n]) + h_{n-1} \circ d_n([g_1|\cdots|g_n]) = [g_1|\cdots|g_n].$$

For $n = 0$, we have

$$\begin{aligned} d_1 \circ h_0([g_1]) &= d_1([e|g_1]) = [g_1] - [e], \\ h_{-1} \circ d_0([g_1]) &= h_{-1}(\epsilon([g_1])) = h_{-1}([e]) = [e]. \end{aligned}$$

Therefore, we have $d_{n+1} \circ h_n + h_{n-1} \circ d_n = \mathrm{id}_{B_n}$ for all $n \geq -1$. □

By the previous theorem, we have a free resolution of $\mathbb{Z}$ as $\mathbb{Z}[G]$-modules. This is called the *bar resolution* of $\mathbb{Z}$ as $\mathbb{Z}[G]$-modules invented by Eilenberg and MacLane.

We also observe that $\mathrm{Hom}_{\mathbb{Z}[G]}(B_n, B_{n-1}) \cong \mathrm{Hom}_{\mathsf{Set}}(G^n, B_{n-1})$ because B_n is a free $\mathbb{Z}[G]$-module on G^n.

Let's consider a special case: $B_2 \xrightarrow{d_2} B_1 \xrightarrow{d_1} B_0 \xrightarrow{d_0} B_{-1} \to 0$. Let A be a G-module. Then we have

$$H^i(G, A) = H^i(\mathrm{Hom}_{\mathsf{Set}}(G^\bullet, A)).$$

For $G^0 = \{e\}$, we have $\mathrm{Hom}_{\mathsf{Set}}(G^0, A) \cong A$. We consider the pullback map $d_1^* : \mathrm{Hom}_{\mathsf{Set}}(G^0, A) \to \mathrm{Hom}_{\mathsf{Set}}(G^1, A)$ of d_1 which is given by $d^1(g) = g - 1$. So we have $H^0(G, A) = \ker(d_1^*) = \{a \in A \ : \ g \cdot a - a = 0\} = A^G$.

We also have $d_2(f)(g_1, g_2) = g_1 \cdot f(g_2) - f(g_1 g_2) + f(g_1)$. Hence $Z^1(G, A) = \ker(d_2^*) = \{f : G \to A \ : \ g_1 \cdot f(g_2) - f(g_1 g_2) + f(g_1) = 0\}$. Such functions f are called *crossed homomorphisms*. In addition we have $B^1(G, A) = \mathrm{im}(d_1^*) = \{f : G \to A \ : \ f(g) = g \cdot a - a \text{ for some } a \in A\}$. Any map in the boundary is also a crossed homomorphism because

$$g_1 \cdot f(g_2) - f(g_1 g_2) + f(g_1) = g_1 \cdot (g_2 \cdot a - a) - (g_1 g_2 \cdot a - a) + (g_1 \cdot a - a) = 0.$$

And such maps are called *principal crossed homomorphisms*. And $H^1(G, A) = Z^1(G, A)/B^1(G, A)$ is the group of crossed homomorphisms modulo principal crossed homomorphisms.

Corollary 8. Let A be a trivial G-module. Then, we have

$$H^0(G, A) \cong A,$$
$$H^1(G, A) \cong \mathrm{Hom}_{\mathsf{group}}(G, A).$$

Proof. We have $Z^1(G, A) = \mathrm{Hom}_{\mathsf{group}}(G, A)$ and $B^1(G, A) = 0$ since A is a trivial G-module. □

We already looked at the case of $A = \mathbb{Z}$ of the following corollary.

Corollary 9. Suppose G is a finite group and A is a torsion-free trivial G-module. Then, we have $H^1(G, A) = 0$.

Theorem 4. Suppose G is a finite group of order m, and let A be a G-module. Then $H^n(G, A)$ is annihilated by m for all $n > 0$, i.e., $H^n(G, A)$ is a $\mathbb{Z}/m\mathbb{Z}$-module for all $n > 0$.

Proof. Our strategy to prove this is to use null-homotopy again. Let $B_\bullet \xrightarrow{\epsilon} \mathbb{Z}$ be the bar resolution of $\mathbb{Z}$. Let's consider the map of the complexes $B_\bullet \to B_\bullet$ given by where $N = \sum_{g \in G} g$ is the norm element. We want to show that this is null-homotopic in positive degrees. Let $\nu_n : B_n \to B_{n+1}$ be the map given by

$$\nu_n([g_1|g_2|\cdots|g_n]) = \epsilon_n \sum_{g \in G} [g_1|\cdots|g_n|g]$$

$$\begin{array}{ccccccc} \cdots & \longrightarrow & B_2 & \longrightarrow & B_1 & \longrightarrow & B_0 \\ & & \downarrow{\scriptstyle m} & & \downarrow{\scriptstyle m} & & \downarrow{\scriptstyle m-N} \\ \cdots & \longrightarrow & B_2 & \longrightarrow & B_1 & \longrightarrow & B_0 \end{array}$$

where $\epsilon_n = (-1)^{n+1}$. We claim that $\nu_{n-1} \circ d_n + d_{n+1} \circ \nu_n = m$ in all positive degrees. Let $\omega = [g_1|\cdots|g_n] \in B_n$. Then, we have

$$\begin{aligned} d_{n+1} \circ \nu_n(\omega) &= \epsilon_n \sum_{g \in G} d_{n+1}([g_1|\cdots|g_n|g]) \\ &= (-1)^{n+1} \sum_{g \in G} (g_1[g_2|\cdots|g_n|g]) \\ &\quad + (-1)^{n+1} \sum_{g \in G} \left(\sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_i g_{i+1}|\cdots|g_n|g] \right) \\ &\quad + (-1)^{n+1} \sum_{g \in G} ((-1)^n [g_1|\cdots|g_n g]) + \sum_{g \in G} ([g_1|\cdots|g_n]), \\ \nu_{n-1} \circ d_n(\omega) &= \nu_{n-1} \left(g_1[g_2|\cdots|g_n] + \sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_i g_{i+1}|\cdots|g_n] + (-1)^n [g_1|\cdots|g_{n-1}] \right) \\ &= (-1)^n \sum_{g \in G} g_1[g_2|\cdots|g_n|g] + (-1)^n \sum_{g \in G} \left(\sum_{i=1}^{n-1} (-1)^i [g_1|\cdots|g_i g_{i+1}|\cdots|g_n|g] \right) \\ &\quad + (-1)^n \sum_{g \in G} ([g_1|g_2|\cdots|g_{n-1}|g]). \end{aligned}$$

We observe that

$$\sum_{g \in G} ([g_1|g_2|\cdots|g_{n-1}|g]) = \sum_{g \in G} ([g_1|\cdots|g_n g])$$

since g runs over all elements of G. Therefore, from the above, we have

$$d_{n+1} \circ \nu_n(\omega) + \nu_{n-1} \circ d_n(\omega) = \sum_{g \in G} [g_1|\cdots|g_n] = m(\omega).$$

For the 0-th degree, we have

$$d_1 \circ \nu_0([]) = d_1\left(-\sum_{g\in G}[g]\right) = -\sum_{g\in G}[g] + \sum_{g\in G}[1] = m - N.$$

Thus, $m \cdot H^n(G, A) = 0$ for all $n > 0$. □

Corollary 10. If A is $\mathbb{Z}[1/m]$-module, then $H^n(G, A) = 0$ for all $n > 0$.

Proof. Since A is a $\mathbb{Z}[1/m]$-module, we have m is an automorphism for A; hence an automorphism for $H^n(G, A)$ for all $n > 0$. But $m \cdot H^n(G, A) = 0$ for all $n > 0$ by the previous theorem. Therefore, $H^n(G, A) = 0$ for all $n > 0$. □

We continue to assume that G is a finite group of order m. By averaging over G, we get a trivial G-module structure on a $\mathbb{Z}[1/m][G]$-module A.

Definition 3. The Reynolds operator $R : A \to A^G$ is defined by

$$R(a) = \frac{1}{m}\sum_{g\in G} g \cdot a.$$

4 Periodic Cohomology

4.1 Group Cohomology of a Cyclic Group

Now we will compute the cohomology of a group in a simpler case, when the group is cyclic. Let G be a cyclic group of order n generated by σ.

Before beginning the computation, we observe the following:

Remark 4.

1. $\mathbb{Z}[\mathbb{Z}] = \mathbb{Z}[t, t^{-1}]$.
2. $\mathbb{Z}[\mathbb{Z}_{>0}] = \mathbb{Z}[t]$.
3. $\mathbb{Z}[\mathbb{Z}/n\mathbb{Z}] = \mathbb{Z}[t]/(t^n - 1)$.

So we now know that the group ring $\mathbb{Z}[G]$ is isomorphic to the quotient ring $\mathbb{Z}[t]/(t^n-1)$ via the map $\sigma \mapsto t$. Let N be the norm element $N = 1+\sigma+\cdots+\sigma^{n-1} \in \mathbb{Z}[G]$. In $\mathbb{Z}[t]/(t^n - 1)$, we have $N = 1 + t + \cdots + t^{n-1}$. So we observe that $N(\sigma - 1)$ is corresponding to $N(t-1) = (1+t+\cdots+t^{n-1})(t-1) = t^n - 1 = 0$ in $\mathbb{Z}[t]/(t^n - 1)$. It follows that $\mathrm{im}(N) \subseteq \ker(\sigma - 1)$ and $\mathrm{im}(\sigma - 1) \subseteq \ker(N)$ when we consider N and $\sigma - 1$ as multiplication maps on $\mathbb{Z}[G]$. Moreover, we have

Proposition 3. $\ker(N) = \mathrm{im}(\sigma - 1)$ and $\ker(\sigma - 1) = \mathrm{im}(N)$.

Proof. Suppose that $a \in \ker(N)$, i.e., $Na = 0$. Let $a(t)$ be the polynomial in $\mathbb{Z}[t]$ corresponding to a. Then, we have $N(t)a(t) = (t^n - 1)b(t)$ for some $b(t) \in \mathbb{Z}[t]$. Since $N(t)(t-1) = t^n - 1$, we have $a(t) = (t-1)b(t)$ in $\mathbb{Z}[t]$ and hence $a \in \mathrm{im}(\sigma - 1)$; hence $\ker(N) = \mathrm{im}(\sigma - 1)$. With the same argument, we can show that $\ker(\sigma - 1) = \mathrm{im}(N)$. □

With the property above, we can construct a free resolution of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module which is more convenient than the bar resolution. We call this the 2-*periodic resolution* of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module.

Corollary 11. The following chain is exact:

$$\cdots \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{N} \mathbb{Z}[G] \xrightarrow{\sigma-1} \mathbb{Z}[G] \xrightarrow{\epsilon} \mathbb{Z} \to 0$$

where ϵ is the augmentation map.

Suppose now that A is a G-module. Let $B^\bullet \xrightarrow{\epsilon} \mathbb{Z}$ be the 2-periodic resolution of $\mathbb{Z}$ as a $\mathbb{Z}[G]$-module. Then, we have

$$\begin{aligned} &H^i(G, A) = H^i(\mathrm{Hom}_{\mathbb{Z}[G]}(B^\bullet, A)), \\ &\mathrm{Hom}_{\mathbb{Z}[G]}(B^\bullet, A) = \cdots \xleftarrow{N} A \xleftarrow{\sigma-1} A \xleftarrow{N} A \xleftarrow{\sigma-1} A. \end{aligned}$$

Thus, we get

$$\begin{aligned} H^0(G, A) &= \ker(\sigma - 1) = A^G, \\ H^{2i+1}(G, A) &= {}_N A/(\sigma - 1)A, \\ H^{2i+2}(G, A) &= A^G/N(A) \end{aligned}$$

where ${}_N A = \{a \in A : Na = 0\}$ and $i \geq 0$. We can see that the cohomology groups of a cyclic group are periodic with period 2. Here are some applications of this computation.

4.2 Hilbert 90

Let L/K be a finite Galois extension of fields with Galois group $G = \mathrm{Gal}(L/K)$. What are the Galois cohomology groups of G with coefficients in L or $L^\times$? For the 0-th cohomology group, we can easily see that $H^0(G, L) = L^G = K$ and $H^0(G, L^\times) = (L^\times)^G = K^\times$. How about the higher cohomology groups?

Theorem 5 (Hilbert's Theorem 90)**.** $H^1(G, L^\times) = 0$.

Before proving this theorem, we prove the following lemma.

Lemma 4 (Linear independence of automorphisms)**.** Let E be a field and $\sigma_1, \cdots, \sigma_n$ be distinct automorphisms of E. Then, for any $c_1, \cdots, c_n \in E$, $\sum_{i=1}^{n} c_i\sigma_i(x) = 0$ for all $x \in E$ implies $c_1 = c_2 = \cdots = c_n = 0$.

Proof. We will prove this by induction on n. We may assume that none of c_i is zero, otherwise we can remove the corresponding σ_i and reduce to a smaller n. If $n = 1$, then we have $c_1\sigma_1(1) = c_1 \neq 0$.

Now we consider the case $n > 1$ and the statement is true for $n-1$, but suppose that $\sum_{i=1}^{n} c_i\sigma_i = 0$ for the sake of contradiction. We can set $c_n = -1$ without loss of generality. Let $x, y \in E$. We have $\sigma_n(x) = \sum_{i=1}^{n-1} c_i\sigma_i(x)$, and $\sigma_n(x)\sigma_n(y) = \sigma_n(xy)$.

The RHS is equal to $\sum_{i=1}^{n-1} c_i\sigma_i(xy) = \sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_i(y)$ while the LHS is equal to $\sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_n(y)$. It follows that

$$0 = \sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_n(y) - \sum_{i=1}^{n-1} c_i\sigma_i(x)\sigma_i(y) = \sum_{i=1}^{n-1} c_i\sigma_i(x)(\sigma_n(y) - \sigma_i(y)).$$

By the induction hypothesis, we have $c_i(\sigma_n(y) - \sigma_i(y)) = 0$ for all i. We also have $c_i \neq 0$ for all i, so we must have $\sigma_n(y) - \sigma_i(y) = 0$ for all i. Recalling that y is arbitrary, we have $\sigma_n = \sigma_i$ for all i which is a contradiction. □

Proof of Theorem 5. Recall that $H^1(G, L^\times)$ is the group of crossed homomorphisms modulo principal crossed homomorphisms. Let $\alpha : G \to L^\times$ be a crossed homomorphism. For $g, h \in G$, we have $\alpha(gh) = g\alpha(h) \cdot \alpha(g)$. We note that each $g \in G$ is an automorphism of $L^\times$, and none of $\alpha(g)$ is zero. The lemma implies that there exists $x \in L^\times$ such that $\beta := \sum_{g\in G} \alpha(g)g(x) \neq 0$. For $h \in G$, we have

$$\begin{aligned} h(\beta) &= h\left(\sum_{g\in G} \alpha(g)g(x)\right) = \sum_{g\in G} h(\alpha(g))hg(x) \\ &= \sum_{g\in G} \alpha(h)^{-1}\alpha(hg)hg(x) = \alpha(h)^{-1}\sum_{g\in G} \alpha(hg)hg(x) = \alpha(h)^{-1}\beta. \end{aligned}$$

So we have $\alpha(h) = \beta/h(\beta)$ for all $h \in G$. Therefore, α is a principal crossed homomorphism, and we conclude that $H^1(G, L^\times) = 0$. □

Let's take a look at the original version of Hilbert 90.

Theorem 6 (Hilbert's Theorem 90, Original Version)**.** Let L/K be a cyclic Galois extension of fields with Galois group $G = \mathrm{Gal}(L/K)$ generated by σ. Let $N : L^\times \to K^\times$ be the norm map given by $N(x) = x\sigma(x)\cdots\sigma^{n-1}(x)$. Then, for any $x \in L^\times$ with $N(x) = 1$, there exists $y \in L^\times$ such that $x = y/\sigma(y)$.

Proof. We want to show that Theorem 5 implies the original version of Hilbert 90. Recall that $H^1(G, L^\times) = \ker(N)/\mathrm{im}(\sigma - 1)$ because G is cyclic. So if $N(x) = 1$, then $x \in \ker(N)$, and hence $x \in \mathrm{im}(\sigma - 1)$ by Theorem 5. Therefore, there exists $y \in L^\times$ such that $x = y/\sigma(y)$. □

4.3 Tate Cohomology

This is the main application of the 2-periodic resolution. Let G be a cyclic group acting on an abelian group A.

Definition 4 (Tate Cohomology)**.** The Tate cohomology groups $\hat{H}^n(G, A)$ are defined as follows:

$$\hat{H}^n(G, A) = \begin{cases} A^G/N(A) & n \text{ is even,} \\ {}_N A/(\sigma - 1)A & n \text{ is odd.} \end{cases}$$

So now we can consider the negative indices. In fact, we can do this for groups that are not necessarily cyclic. We define the negative part with the interaction of cohomology and homology.

5 Group Cohomology and Subgroup

When L/K is a finite Galois extension with group G, we have $H^1(G, L^\times) = 0$. What about $H^1(G, L)$? Before we answer this question, we take a look at general theory of cohomology groups.

5.1 Induced and Coinduced Modules

Let's consider more general setting. Let $\alpha : R \to S$ be a ring homomorphism, and they don't have to be commutative. Suppose that S is a projective(or free) R-module via α. Let M be an R-modules, and let N, P be S-modules(R-modules as well), and let $h : M \to N$ be an R-linear map.

Definition 5 (Induced and Coinduced Modules)**.**

1. The *induced S-module* $\mathrm{Ind}_R^S(M)$ is defined as $S \otimes_R M$.
2. The *coinduced S-module* $\mathrm{Coind}_R^S(M)$ is defined as $\mathrm{Hom}_R(S, M)$. We note that for $f \in \mathrm{Hom}_R(S, N)$ and $s \in S$, we define $(s \cdot f)(t) = f(st)$ for $t \in S$.

Proposition 4. There is a canonical isomorphism

$$\mathrm{Ext}_S^i(N, \mathrm{Hom}_R(S, M)) \cong \mathrm{Ext}_R^i(N, M).$$

Proof. Let $P^\bullet \to N$ be a projective(or free resp.) resolution of P as an S-module. Then it is also a projective resolution of N as an R-module since S is a projective R-module. So we have

$$\begin{aligned}
\mathrm{Ext}_R^i(N, M) &\cong H^i(\mathrm{Hom}_R(P^\bullet, M)) \\
&\cong H^i(\mathrm{Hom}_S(P^\bullet, \mathrm{Hom}_R(S, M))) \\
&\cong \mathrm{Ext}_S^i(N, \mathrm{Hom}_R(S, M)).
\end{aligned}$$

□

5.2 Group algebra and subgroup algebra

Lemma 5. Suppose that G is a discrete group and $H \hookrightarrow G$ is a subgroup of finite index. Then we have that $\mathbb{Z}[G]$ is a free $\mathbb{Z}[H]$-module with basis G/H.

Proof. We know that $\mathbb{Z}[H]$ is a free $\mathbb{Z}$-module with basis H, and $\mathbb{Z}[G]$ is a free $\mathbb{Z}$-module with basis G. Let $g_0 = e, g_1, \ldots, g_{n-1}$ be a set of representatives where $n = [G : H]$.

Consider a map $\Phi : \bigoplus_{i=0}^{n-1} \mathbb{Z}[H]\tilde{g_i} \to \mathbb{Z}[G]$ defined by $(h_0, h_1, \ldots, h_{n-1}) \mapsto \sum_{i=0}^{n-1} g_i h_i$. It is clear Φ is surjective since every $g = hg_i$ for some $h \in H$. We claim that Φ is injective as well. With $g \mapsto h\tilde{g_i}$, we get a map $\Psi : \mathbb{Z}[G] \to \bigoplus_{i=0}^{n-1} \mathbb{Z}[H]\tilde{g_i}$. We notice that $\Phi \circ \Psi = \mathrm{id}_{\mathbb{Z}[G]}$, so Φ is injective. □

So we can consider $R = \mathbb{Z}[H]$ and $S = \mathbb{Z}[G]$ in the previous subsection. We can consider the induced and coinduced module maps: $\mathrm{Ind}_H^G := \mathrm{Ind}_R^S$ and $\mathrm{Coind}_H^G := \mathrm{Coind}_R^S$. We can set M to be an H-module, and N to be $\mathbb{Z}$ as a trivial G-module. So the following proposition is a direct consequence of the previous proposition.

Proposition 5 (Shapiro's Lemma)**.**

$$\mathrm{Ext}_S^i(\mathbb{Z}, \mathrm{Coind}_H^G(A)) \cong \mathrm{Ext}_R^i(\mathbb{Z}, A).$$

Corollary 12. Let G be a finite group and A be any abelian group. Then we have $H^i(G, \mathrm{Coind}_*^G(A)) = A$ if $i = 0$ and 0 otherwise where $\mathrm{Coind}_*^G(A) = \mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], A)$.

Proof. It follows from Shapiro's lemma when $H = \{e\}$. □

5.3 More Discussion on Hilbert 90

Theorem 7. Let L/K be a finite Galois extension with group G. Then we have $H^0(G, L) = K$ and $H^i(G, L) = 0$ for all $i > 0$.

Proof. By normal basis theorem, we have some $x \in L$ such that $\{g(x)\}_{g \in G}$ is a basis of L over K. We observe $\mathrm{Hom}_{\mathbb{Z}}(\mathbb{Z}[G], K) \approx \mathbb{Z}[G] \underset{\mathbb{Z}}{\otimes} K$ and $K[G] \cong L$ since $\mathbb{Z}[G]$

is a free $\mathbb{Z}$-module. Hence $\mathrm{Coind}_*^G(K) \cong L$. From the previous corollary, we have $H^i(G, L) = H^i(G, \mathrm{Coind}_*^G(K)) = K$ if $i = 0$ and 0 otherwise. $\square$

Corollary 13. If L/K is a cyclic extension with group $G = \langle \sigma \rangle$, then we have an exact sequence $0 \to K \to L \xrightarrow{\sigma - 1} L \xrightarrow{\mathrm{tr}_{L/K}} K \to 0$.

Proof. The exactness at the first L is clear since $\sigma(x) = x$ if and only if $x \in K$. We have $0 = H^1(G, L) = {}_N L/(\sigma - 1)L$, and the norm map is exactly the trace map $\mathrm{tr}_{L/K}$; hence we have the exactness at the second L. We also have $0 = H^2(G, L) = L^G/NL = K/\mathrm{tr}_{L/K}(L)$; we have the exactness at the last K. $\square$

6 Change of Groups

Suppose that G is a group acting on an abelian group A, and suppose that there is a group homomorphism $\phi : G' \to G$. Then ϕ induces a G'-action on A.

Question 1. Any connection between $H^i(G, A)$ and $H^i(G', A)$?

For $i = 0$, we have $H^0(G, A) = A^G$ and $H^0(G', A) = A^{G'}$, and we have an inclusion $A^G \hookrightarrow A^{G'}$.

6.1 Change of Rings

We consider a more general setting as before. Let $\alpha : R \to S$ be a ring homomorphism and they don't have to be commutative. We are given S-modules P, M and an R-module N. We have an R-linear map $h : M \to N$.

Theorem 8. Given the above data, there is a natural abelian group homomorphism $h_* : \mathrm{Ext}^i_S(P, M) \to \mathrm{Ext}^i_R(P, N)$ for all $i \geq 0$. Moreover, if R is commutative, then h_* is an R-linear map.

Proof. Let $M \to I^\bullet$ be an injective resolution of M as an S-module. Let $I^\bullet \to J^\bullet$ be an injective resolution of $I^\bullet$ as an R-module. Then we have canonically induced maps:

$$\begin{array}{ccc} \mathrm{Hom}_S(P, M) & \longrightarrow & \mathrm{Hom}_R(P, I^\bullet) \\ & & \downarrow{\scriptstyle \alpha^*} \\ & & \mathrm{Hom}_R(P, I^\bullet) \\ & & \downarrow \\ & & \mathrm{Hom}_R(P, J^\bullet) \end{array}$$

hence we have $H^i(\mathrm{Hom}_S(P, I^\bullet))(= \mathrm{Ext}^i_S(P, M)) \to H^i(\mathrm{Hom}_R(P, J^\bullet))(= \mathrm{Ext}^i_R(P, M))$. With $h : M \to N$, a map $\mathrm{Ext}^i_R(P, M) \to \mathrm{Ext}^i_R(P, N)$ is induced. Thus we have a map $h_* : \mathrm{Ext}^i_S(P, M) \to \mathrm{Ext}^i_R(P, N)$ by composition. □

We note that h_* is a compusition of $\mathrm{Hom}_S(P, M) \hookrightarrow \mathrm{Hom}_R(P, M)$ and $h \circ (-) : \mathrm{Hom}_R(P, M) \to \mathrm{Hom}_R(P, N)$ when $i = 0$.

6.2 Restriction Maps

Proposition 6. Let $\varphi : G' \to G$ be a group homomorphism and A be a G-module. Then we have a natural map $\varphi^* : H^i(G, A) \to H^i(G', A)$ for all $i \geq 0$.

Proof. Take $R = \mathbb{Z}[G']$, $S = \mathbb{Z}[G]$, $P = \mathbb{Z}$, $M = A$, and $N = B$. Then we have a natural map $\varphi^* : \mathrm{Ext}^i_{\mathbb{Z}[G]}(\mathbb{Z}, A) \to \mathrm{Ext}^i_{\mathbb{Z}[G']}(\mathbb{Z}, B)$ for all $i \geq 0$. □

We note that when $i = 0$, we have $A^G \hookrightarrow A^{G'} \overset{h_*}{\hookrightarrow} B^{G'}$.

Example 1 (Restriction Maps)**.**

1. Take $A = B$ and $h = \mathrm{id}_A$. Then we have a natural map $\mathrm{Res}^G_{G'} : H^i(G, A) \to H^i(G', A)$ for all $i \geq 0$. As we have seen, when $i = 0$, we have $A^G \hookrightarrow A^{G'}$.
2. Take G' to be a subgroup H of G. Then we get a restriction map $\mathrm{Res}^G_H : H^i(G, -) \to H^i(H, -)$ on G-modules.

Example 2 (Topological Interpretation of Restriction Maps)**.** Let H be a subgroup of G. Recall from Section 1 that $H^i(G, \mathbb{Z}) \cong H^i_{sing}(BG, \mathbb{Z})$. So the restriction map Res^G_H should be a map

$$H^*_{sing}(BG, \mathbb{Z}) \to H^*_{sing}(BH, \mathbb{Z}),$$

and a natural way to produce such a map in topology is to pull back along a continuous map.

Question 2. Is there a continuous map $BH \to BG$?

The tool is the universal property of BG which classifies principal G-bundles. That is, for a (paracompact) space X,

$$[X, BG] \xrightarrow{\sim} \{\text{principal } G\text{-bundles over } X\}/\cong, \qquad [f] \mapsto f^*EG.$$

Hence to construct a map $BH \to BG$ (up to homotopy), it is enough to construct a principal G-bundle over BH. Given a principal H-bundle $EH \to BH$, we want to

enlarge its structure group from H to G. We can do this by taking the associated bundle $EH \times^H G$ with fiber G: $EH \overset{H}{\times} G \to BH$, $[e, g] \mapsto [e]$.

By the universal property, there is a continuous map $\iota : BH \to BG$, unique up to homotopy, such that $\iota^* EG \cong EH \overset{H}{\times} G$ as principal G-bundles. Here are some remarks on this construction.

1. A concrete model: G acts freely on the contractible space EG, hence so does H. So EG is also a model for EH, and $BH = EG/H$. In this model ι is simply the quotient map

$$BH = EG/H \longrightarrow EG/G = BG,$$

   which is a covering map with fiber G/H (of degree $[G : H]$ when it is finite).

2. The map ι induces $\iota^* : H^*_{sing}(BG, \mathbb{Z}) \to H^*_{sing}(BH, \mathbb{Z})$, and under the identifications $H^*(G, \mathbb{Z}) \cong H^*_{sing}(BG, \mathbb{Z})$, $H^*(H, \mathbb{Z}) \cong H^*_{sing}(BH, \mathbb{Z})$ this is the restriction map Res^G_H. Indeed, on π_1 the map ι is the inclusion $H \hookrightarrow G$, and Res^G_H is the map on cohomology induced by this group homomorphism.

6.3 Inflication Maps

Suppose that H is a normal subgroup of G, so we can consider the quotient map $\pi : G \to G/H$.

Let $\alpha : R \to S$ be a ring homomorphism such that S is a free-R-module of rank $n < \infty$. Let M, P be S-modules. Then we have the corestriction map $\mathrm{Cores}_R^S : \mathrm{Ext}_R^i(P, M) \to \mathrm{Ext}_S^i(P, M)$ and the restriction map $\mathrm{Res}_R^S : \mathrm{Ext}_S^i(P, M) \to \mathrm{Ext}_R^i(P, M)$.

Proposition 7. The composition $\mathrm{Cores}_R^S \circ \mathrm{Res}_R^S$ is multiplication by n.

Proof. First, we start with the case $i = 0$. We have the following diagram:

$$\begin{array}{ccccc} \mathrm{Hom}_S(P, M) & \xhookrightarrow{\mathrm{Res}_R^S} & \mathrm{Hom}_R(P, M) & \xrightarrow{\sim} & \mathrm{Hom}_S(P \underset{R}{\otimes} S, M) \\ & & & & \downarrow \wr \\ \mathrm{Hom}_S(P, M) & \xleftarrow{\Delta^*} & \bigoplus \mathrm{Hom}_S(Pe_i, M) & \xleftarrow{\sim} & \mathrm{Hom}_S(P^n, M) \end{array}$$

where $e_1, \ldots, e_n$ is a basis of S as a free-R-module. We have $\Delta^*(f_1, \ldots, f_n) = \sum_{i=1}^n f_i$. We observe that the composition gives $f \mapsto (f, \ldots, f) \mapsto \sum_{i=1}^n f = nf$.

It gives a natural transformation $\eta : \mathrm{Hom}_S(P, -) \to \mathrm{Hom}_S(P, -)$ on S-Mod such that $\eta(M)$ is multiplication by n. Since $\mathrm{Hom}_S(P, -)$ is a left exact functor, we can extend η to a natural transformation $\eta^i : \mathrm{Ext}_S^i(P, -) \to \mathrm{Ext}_S^i(P, -)$ for all $i \geq 0$, and it must be multiplication by n as well. □

Corollary 14. Let G be a discrete group and H be a subgroup of a finite index n. Let $A \in G$-Mod. Then the composition

$$H^i(G, A) \xrightarrow{\mathrm{Res}_H^G} H^i(H, A) \xrightarrow{\mathrm{Cores}_H^G} H^i(G, A)$$

is multiplication by $n = |G/H|$.

This can be interpreted as the algebraic version of a topological fact. Let X be a connected topological space and $Y \to X$ be an n-sheeted covering space. Then the restriction map is corresponding to the pullback p^* and the corestriction map is corresponding to the transfer map $p_!$. (So we sometimes call the corestriction map the transfer map.) Copying a class to n sheets and adding them back up gives n times the original class.

Corollary 15. Suppose G is a finite group of order n and A is a G-module. Then $H^i(G, A)$ is annihilated by n for all $i > 0$.

Proof. Apply the previous corollary with the trivial subgroup $H = \{e\}$. Then we observe $H^i(G, A) \to H^i(\{e\}, A) \to H^i(G, A)$ is multiplication by n. But $H^i(\{e\}, A) = 0$ for all $i > 0$, so we conclude that $H^i(G, A)$ is annihilated by n for all $i > 0$. □

Let G be a discrete group and H be a normal subgroup of finite index, and A be a G-module. Then we have the inflation map, the

7 Group Cohomology for Profinite Groups

7.1 Topological Groups

Definition 6 (Topological Group)**.**

1. A *topological group* is a group G equipped with a topology such that the multiplication map $G \times G \to G$, $(g, h) \mapsto gh$ and the inversion map $G \to G$, $g \mapsto g^{-1}$ are continuous.

2. Let G and H be topological groups. A *topological group homomorphism* is a group homomorphism $\varphi : G \to H$ which is continuous. Two topological groups G and H are *isomorphic* if they are isomorphic as groups and homeomorphic as topological spaces.

Example 3 (Examples of Topological Groups)**.**

1. Every group is a topological group with the discrete topology.

2. There are classical groups such as $\mathrm{GL}(n, \mathbb{R})$ and $\mathrm{GL}(n, \mathbb{C})$.

3. The p-adic Lie groups such as $\mathrm{GL}(n, \mathbb{Q}_p)$ and $\mathrm{GL}(n, \mathbb{Z}_p)$ are topological groups with the p-adic topology. Unlike the classical groups, these groups are totally disconnected.

Here are some useful facts about topological groups.

Theorem 9. G is metrizable if and only if it is Hausdorff and first countable.

Theorem 10. Let G be a topological group and H be a subgroup of G. Then the closure $\overline{H}$ of H is also a subgroup of G. Moreover, if H is normal, then $\overline{H}$ is also normal.

Theorem 11. Let G be a Hausdorff topological group and H be a discrete subgroup of G. Then H is closed in G.

Proof. Since H is discrete, we have $\{e\}$ is open in H so that we can write $\{e\} = H \cap U$ for some open set U in G. Since $(x, y) \mapsto x^{-1}y$ is continuous, we have $V \subset G$ open such that $e \in V$ and $V^{-1}V \subset U$.

We claim that $gV \cap H$ has at most one element for any $g \in G$. Suppose that $h_1, h_2 \in gV \cap H$ and we write $h_1 = gv_1$ and $h_2 = gv_2$ for some $v_1, v_2 \in V$. Then we have $h_1^{-1}h_2 = v_1^{-1}g^{-1}gv_2 = v_1^{-1}v_2 \in V^{-1}V \subset U$. It implies that $h_1^{-1}h_2 \in H \cap U = \{e\}$, so we have $h_1 = h_2$.

Now we take $g \in \overline{H}$. Since gV is an open neighborhood of g, we have $gV \cap H \neq \emptyset$. By the claim, we have $gV \cap H = \{h\}$. If $g \neq h$, we have a neighborhood W which doesn't have h. Then $gV \cap W$ is an open neighborhood of g which doesn't have any element in H, contradicting that $g \in \overline{H}$. Hence g must be equal to $h \in H$, so we have $\overline{H} = H$. □

7.2 Profinite Group

Definition 7 (Profinite Group)**.** A *profinite group* is a topological group which has an inverse system of finite groups $\{G_i\}_{i \in I}$ such that G is isomorphic to the inverse limit $\varprojlim_{i \in I} G_i$ as topological groups where the inverse system is equipped with the transition maps $\varphi_{ij} : G_j \to G_i$ for $i \leq j$. The *inverse limit* of $\{G_i\}_{i \in I}$ is a topological space

$$\varprojlim_{i \in I} G_i := \left\{ (g_i)_{i \in I} \in \prod_{i \in I} G_i : g_i = \varphi_{ij}(g_j) \text{ for all } i \leq j \right\}$$

with the subspace topology from the product topology on $\prod_{i \in I} G_i$.

We note that the discreteness of G_i does not imply the discreteness of G.

Proposition 8 (Inverse limit is always closed.)**.** G is closed in $G' = \prod_{i \in I} G_i$.

Proof. Let $g = (g_i)_{i \in I} \in G' \setminus G$. Then there exists $\alpha < \beta$ such that $g_\alpha \neq \varphi_{\alpha\beta}(g_\beta)$. Take $U_{\alpha\beta} = \{g_\alpha\} \times \{g_\beta\} \times \prod_{i \in I \setminus \{\alpha, \beta\}} G_i$. Then $U_{\alpha\beta}$ is an open neighborhood of g in G' and $U \cap G = \emptyset$. We notice that $G' \setminus G$ is the union of all such $U_{\alpha\beta}$, so it is open in G'. Thus G is closed in G'. □

Proposition 9. Let $G \hookrightarrow G'$ be as above.

1. G is closed in G'.
2. G is Hausdorff.
3. G is compact.
4. G is totally disconnected i.e. the connected components of G are singletons.

Proof. The first statement is proved in the previous proposition. Each G_i is Hausdorff, so G' is Hausdorff. Since G is closed in G', it is also Hausdorff. Each G_i is compact, so G' is compact by Tychonoff's theorem. So is G since it is closed in G'.

Let $S \subset G$ be a connected component. Then S is connected in G' since G is closed in G'. □